AP & RAP Assignment 2

1. Let $(X, \mathcal{A})$ be a measurable space. Let μ, ν be two measures on $(X, \mathcal{A})$. Show that for all $a, b \in [0, \infty)$ the set-function $\rho(A) := a\mu(A) + b\nu(A), A \in \mathcal{A}$, is again a measure.

2. Let $(\Omega, \mathcal{A}, P)$ be a probability space, i.e., P is a probability measure on the measurable space $(\Omega, \mathcal{A})$. Suppose $(A_j)_{j\in\mathbb{N}} \subset \mathcal{A}$ is a sequence of sets with $P(A_j) = 1$ for all $j \in \mathbb{N}$. Show that $P\left(\bigcap_{j=1}^{\infty} A_j\right) = 1$.

3. Let $(X, \mathcal{A}, \mu)$ be a finite measure space and $(A_j)_{j\in\mathbb{N}}, (B_j)_{j\in\mathbb{N}} \subset \mathcal{A}$ such that $A_j \supset B_j$ for all $j \in \mathbb{N}$. Show that

$$\mu\left(\bigcup_{j=1}^{\infty} A_j\right) - \mu\left(\bigcup_{j=1}^{\infty} B_j\right) \leqslant \sum_{j=1}^{\infty} \left(\mu(A_j) - \mu(B_j)\right)$$

4. Let $(\Omega, \mathcal{A})$ be a measurable space, and let $\mu : \mathcal{A} \to [0, \infty]$ be a mapping that satisfies the following conditions: $\mu(\emptyset) = 0$ and $\mu(A \cup B) = \mu(A) + \mu(B)$ for any disjoint sets $A, B \in \mathcal{A}$, and for any sequence $(A_n) \subset \mathcal{A}$ with $A_n \uparrow A$, we have $\lim_{n\to\infty} \mu(A_n) = \mu(A)$. Prove that μ is a measure.